Lecture 24

Saturday, May 9, 2026 9:04 AM

Today: Principal Component Analysis (PCA)

- Recap of key linear algebra concepts
- What is the goal of PCA?
- PCA as an optimization problem & its solution
- Examples

$$\rightarrow \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix}$$

$$a \det\left(\begin{bmatrix} f & g & h \\ j & k & l \\ n & o & p \end{bmatrix}\right) - b \det\left(\begin{bmatrix} e & g & h \\ i & k & l \\ m & o & p \end{bmatrix}\right)$$

$$+ c \det\left(\begin{bmatrix} e & f & h \\ i & j & l \\ m & n & p \end{bmatrix}\right) - d \det\left(\begin{bmatrix} e & f & g \\ i & j & k \\ m & n & o \end{bmatrix}\right)$$

Computing eigenvalues or eigenvector

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \qquad \lambda$$

$$Av = \lambda v \quad — (1)$$

* tricky

$$RHS = \lambda \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \lambda v_1 \\ \lambda v_2 \end{bmatrix} = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$= \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$(1) \Rightarrow (A - \lambda I) v = 0$$

$$\Rightarrow \begin{pmatrix} a-\lambda & b \\ c & d-\lambda \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \quad — (2)$$

(2) is possible only if $\det \begin{pmatrix} a-\lambda & b \\ c & d-\lambda \end{pmatrix} = 0$

$$\Rightarrow (a-\lambda)(d-\lambda) - bc = 0$$

solve this to obtain λ, use λ to obtain v

$A : m \times m \Rightarrow$ polynomial of degree m

$\Rightarrow$ at most m roots

PCA

n sample points, each is d dimension

$d = 10^4$ $\qquad$ p $\quad p \ll d$

$$\begin{array}{c} d \\ \begin{bmatrix} \text{---}\ x^{(1)}\ \text{---} \\ \text{---}\ x^{(2)}\ \text{---} \\ \vdots \\ \text{---}\ x^{(n)}\ \text{---} \end{bmatrix}_{n \times d} \end{array} \longrightarrow \begin{array}{c} p \\ \begin{bmatrix} \text{---}\ \tilde{x}^{(1)}\ \text{---} \\ \text{---}\ \tilde{x}^{(2)}\ \text{---} \\ \vdots \\ \text{---}\ \tilde{x}^{(n)}\ \text{---} \end{bmatrix}_{n \times p} \end{array}$$

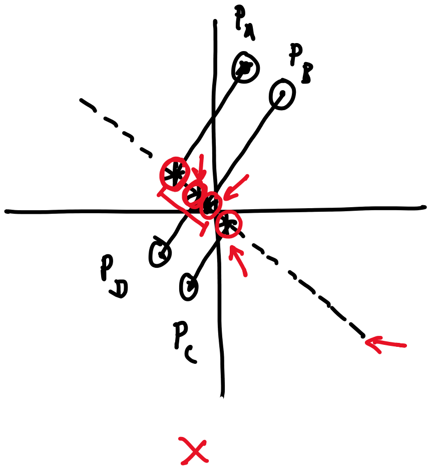
inches $\qquad$ cm

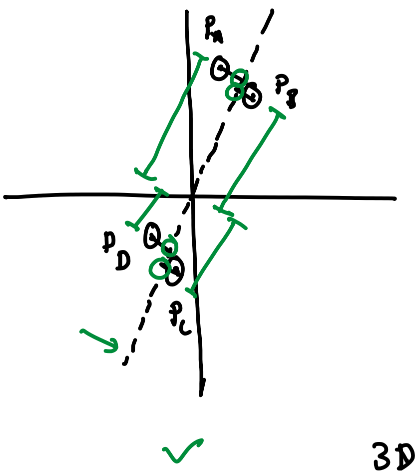
- computation / storage
- visualization
- redundancy

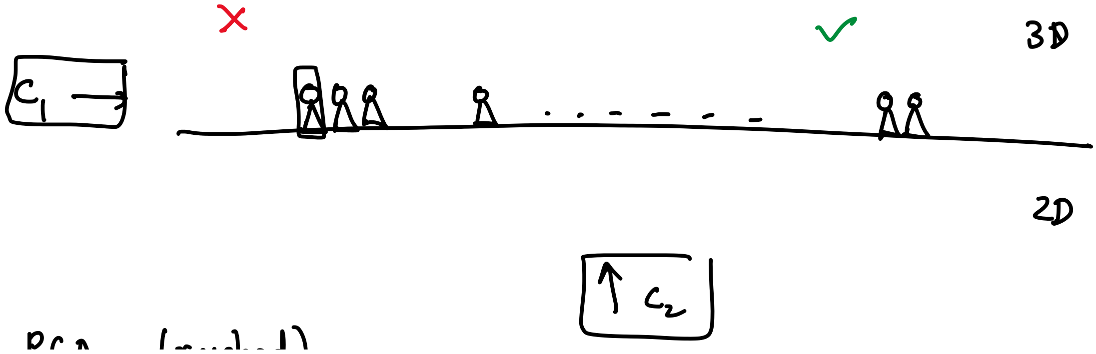
$d = 2 \quad p = 1$

$\rightarrow$ PCA : Transform data from $\mathbb{R}^d$ to $\mathbb{R}^p$ such that the variance in the data is maximized.

$n = 4$







PCA [illegible]

PCA (Pushed)

- n data points $x_1 \dots x_n \in \mathbb{R}^d$
- maximize the variance of the projection

$$\max_{u_1} \underbrace{u_1^T S u_1}_{S \,=\, \text{covariance matrix of the data}} \quad \text{s.t.} \quad \underbrace{\|u_1\| = 1}_{u_1 \text{ is unit vector}}$$

- When you solve the problem (spectral ~~stu~~ theorem)

  u_1 – eigenvector of $\underline{\underline{S}}$ corresponding to the largest eigenvalue

Proof omitted